Scouting America

Plant Science Merit Badge Workbook - Field Botany

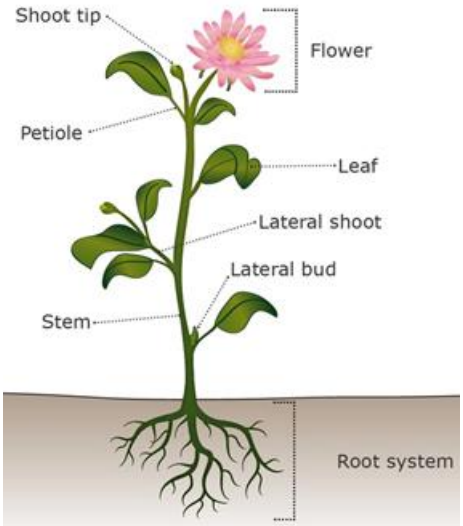
SCOUTMASTER BUCKY

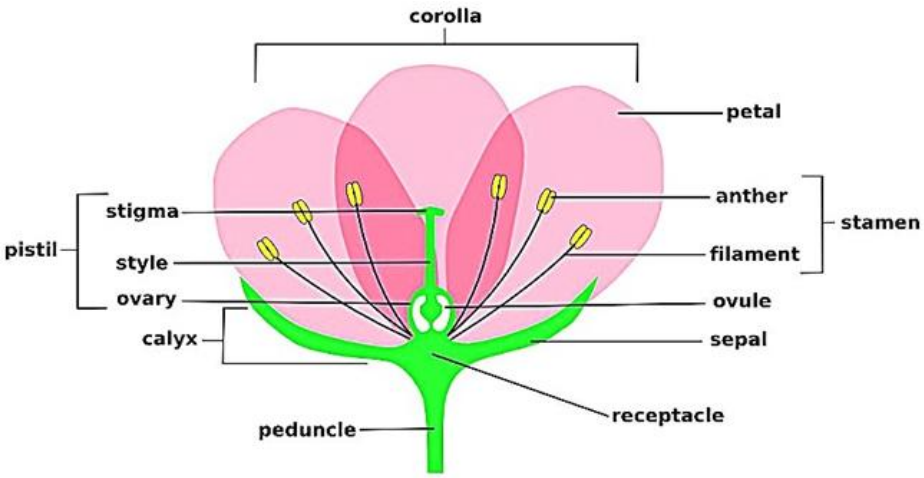
Scouts participating in a Scoutmaster Bucky merit badge opportunity, whether online or in person, should consider using the Plant Science merit badge pamphlet for discovery and knowledge, along with the class preparation pages for clarifications, insights, and expectations.

https://scoutmasterbucky.com/merit-badges/plant-science/plant-science-pamphlet.pdf

https://scoutmasterbucky.com/merit-badges/plant-science/plant-science-cpp.pdf

REQUIREMENT 1: Make a drawing and identify five or more parts of a flowering plant. Tell what each part does.





Notes:

REQUIREMENT 2:	Explain photosynthesis.
Notes:	

REQUIREMENT 2:	Tell why photosynthesis is important.
Notes:	

REQUIREMENT 2:	Tell at least five ways that humans depend on plants.
Way #1:	
Way #2:	
Way #3:	
Way #4:	
Way #5:	

REQUIREMENT 3:	Explain how honeybees and other pollinating insects are important to plant life.
Notes:	

REQUIREMENT 4:	Explain how water affects plants.
Notes:	

REQUIREMENT 4:	Explain how light affects plants.
Notes:	

REQUIREMENT 4:	Explain how air affects plants.
Notes:	

REQUIREMENT 4:	Explain how temperature affects plants.
Notes:	

REQUIREMENT 4:	Explain how pests affect plants.
Notes:	

REQUIREMENT 4:	Describe the nature and function of soil.
Notes:	

REQUIREMENT 4:	Explain the importance of soil.
Notes:	

REQUIREMENT 4:	Tell about the texture of fertile soil.
Notes:	

REQUIREMENT 4:	Tell about the structure of fertile soil.
Notes:	

REQUIREMENT 4:	Tell about the composition of fertile soil.
Notes:	

REQUIREMENT 4:	Tell how soil may be improved.
Notes:	

REQUIREMENT 5:	Tell how to propagate plants by seeds.
Notes:	

REQUIREMENT 5:	Tell how to propagate plants by roots.
Notes:	

REQUIREMENT 5:	Tell how to propagate plants by cuttings.
Notes:	

REQUIREMENT 5:	Tell how to propagate plants by tubers.
Notes:	

REQUIREMENT 5:	Tell how to propagate plants by grafting.
Notes:	

REQUIREMENT 5:	Grow a plant by ONE of these methods (seeds, roots, cuttings, tubers, or grafting).
Be sure to bring proof of your plant project to share your work with your counselor **This requirement must be reviewed with your merit badge counselor.** **BE PREPARED!**	
Notes:	

REQUIREMENT 6:	List by common name at least 10 native plants that grow near your home.
NATIVE PLANT #1:	
Common Name:	
NATIVE PLANT #2:	
Common Name:	
NATIVE PLANT #3:	
Common Name:	
NATIVE PLANT #4:	
Common Name:	
NATIVE PLANT #5:	
Common Name:	
NATIVE PLANT #6:	
Common Name:	
NATIVE PLANT #7:	
Common Name:	
NATIVE PLANT #8:	
Common Name:	
NATIVE PLANT #9:	
Common Name:	
NATIVE PLANT #10:	
Common Name:	

REQUIREMENT 6:	List by common name at least 10 cultivated plants that grow near your home.
CULTIVATED PLANT #1:	
Common Name:	
CULTIVATED PLANT #2:	
Common Name:	
CULTIVATED PLANT #3:	
Common Name:	
CULTIVATED PLANT #4:	
Common Name:	
CULTIVATED PLANT #5:	
Common Name:	
CULTIVATED PLANT #6:	
Common Name:	
CULTIVATED PLANT #7:	
Common Name:	
CULTIVATED PLANT #8:	
Common Name:	
CULTIVATED PLANT #9:	
Common Name:	
CULTIVATED PLANT #10:	
Common Name:	

REQUIREMENT 6: List five invasive non-native plants in your area and tell how they may be harmful.
INVASIVE NON-NATIVE PLANT #1:
Name:
How they may be harmful:
INVASIVE NON-NATIVE PLANT #2:
Name:
How they may be harmful:
INVASIVE NON-NATIVE PLANT #3:
Name:
How they may be harmful:

INVASIVE NON-NATIVE PLANT #4:
Name:
How they may be harmful:
INVASIVE NON-NATIVE PLANT #5:
Name:
How they may be harmful:
REQUIREMENT 6: Tell how the spread of invasive plants may be avoided.
Notes:

REQUIREMENT 6:	Tell how the spread of invasive plants may be controlled in ways that are not damaging to humans.
Notes:	

REQUIREMENT 6:	Tell how the spread of invasive plants may be controlled in ways that are not damaging to wildlife.
Notes:	

REQUIREMENT 6:	Tell how the spread of invasive plants may be controlled in ways that are not damaging to the environment.
Notes:	

REQUIREMENT 7:	Name and tell about careers in agronomy.
Notes:	

REQUIREMENT 7:	Name and tell about careers in horticulture.
Notes:	

REQUIREMENT 7:	Name and tell about careers in botany.
Notes:	

REQUIREMENT 7:	Write a paragraph about a career in one of these fields that interests you.
Selected Field:	
Notes:	

Requirement 8 - Field Botany Option

REQUIREMENT 8f4a REQUIRES PARENT / GUARDIAN PERMISSION.

FIELD BOTANY OPTION REQUIREMENT 8a:	Visit a park, forest, Scout camp, or other natural area near your home.
Place Name and Location:	
FIELD BOTANY OPTION REQUIREMENT 8a1:	Determine which species of plants are the largest and which are the most abundant. Note whether they cast shade on other plants.
Notes:	

FIELD BOTANY OPTION REQUIREMENT 8a2:	Using information from maps, textbooks, or the internet, describe the environmental factors that may influence the presence of plants on your site, including latitude, climate, air and soil temperature, soil type and pH, geology, hydrology, and topography.
Latitude:	
Climate:	
Air Temperature:	
Soil Temperature:	

Soil Type and pH:
Geology:
Hydrology:
Topography:

FIELD BOTANY OPTION REQUIREMENT 8a3:	Record any differences in the types of plants you see at the edge of a forest, near water, in burned areas, or near a road or railroad.
Edge of Forest:	
Near Water:	
In Burned Areas:	
Near a Road or Railroad:	

FIELD BOTANY OPTION REQUIREMENT 8b:	Select a study site that is at least 100 by 100 feet. Make a list of 10 woody plants (trees and shrubs) and 10 non-woody plants in the study site. Find out which of these are native plants and which are exotic (or nonnative).

Study Site Name and Location:

Date Visited:

PLANT IDENTIFICATION CHECK LIST

	Canopy Tree	Small Tree	Shrub	Wildflower / Grass	Vine	Fern	Moss	Algae	Fungus	Lichen	native	nonnative
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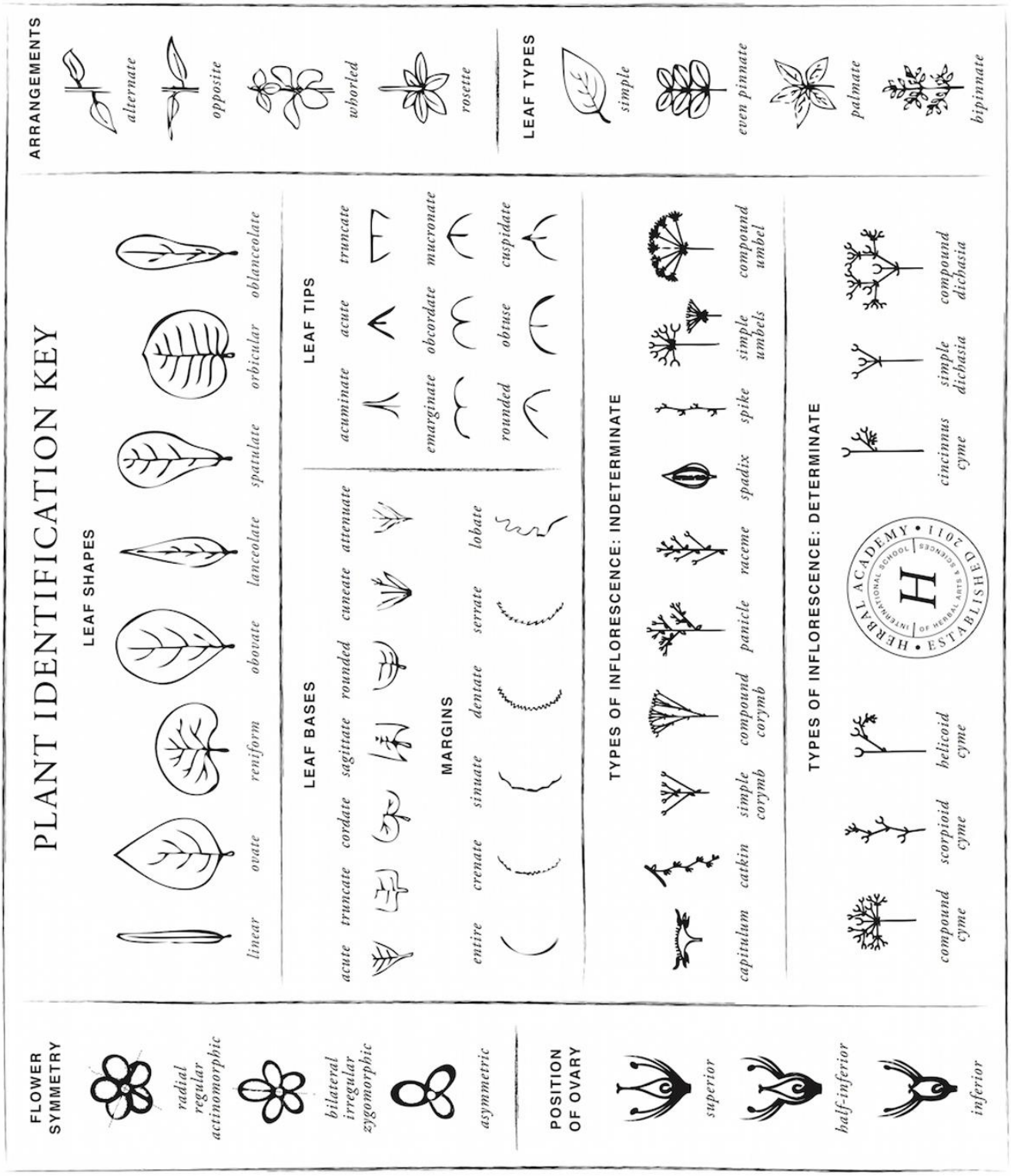
FIELD BOTANY OPTION REQUIREMENT 8c:	Tell how an identification key works and use a simple key to identify 10 kinds of plants (in addition to those in general requirement 5 above).



Notes:

FIELD BOTANY OPTION REQUIREMENT 8c:	Tell the difference between common and scientific names.

Notes:

FIELD BOTANY OPTION REQUIREMENT 8c:	Tell why scientific names are important.

Notes:

FIELD BOTANY OPTION REQUIREMENT 8d:	After gaining permission, collect, identify, press, mount, and label 10 different plants that are common in your area.
This requirement must be reviewed with your merit badge counselor. **BE PREPARED!**	
FIELD BOTANY OPTION REQUIREMENT 8d:	Tell why voucher specimens are important for documentation of a field botanist's discoveries.
This requirement must be reviewed with your merit badge counselor. **BE PREPARED!**	
Notes:	
FIELD BOTANY OPTION REQUIREMENT 8e:	Obtain a list of rare plants of your state.
Notes:	

FIELD BOTANY OPTION REQUIREMENT 8e:	Tell what is being done to protect rare plants and natural areas in your state.

Notes:

FIELD BOTANY OPTION REQUIREMENT 8e:	Write a paragraph about one of the rare plants in your state.

Notes:

DO ONE OF THE FOLLOWING (8F1, 8F2, 8F3, 8F4, OR 8F5) FOR REQUIREMENT 8F IN THE FIELD BOTANY OPTION

FIELD BOTANY OPTION REQUIREMENT 8f1a:	TREE INVENTORY - Identify the trees of your neighborhood, a park, a section of your town, or a Scout camp.
Place Name and Location:	
FIELD BOTANY OPTION REQUIREMENT 8f1b:	TREE INVENTORY - Collect, press, and label leaves, flowers, or fruits to document your inventory.
This requirement must be reviewed with your merit badge counselor. **BE PREPARED!**	
FIELD BOTANY OPTION REQUIREMENT 8f1c:	TREE INVENTORY - List the types of trees by scientific name and give common names. Note the number and size (diameter at 4 feet above ground) of trees observed and determine the largest of each species in your study area.
TREE #1:	
Common Name:	
Scientific Name:	
Notes:	
TREE #2:	
Common Name:	
Scientific Name:	
Notes:	

TREE #3:
Common Name:
Scientific Name:
Notes:

TREE #4:
Common Name:
Scientific Name:
Notes:

TREE #5:
Common Name:
Scientific Name:
Notes:

TREE #6:
Common Name:
Scientific Name:
Notes:

TREE #7:
Common Name:
Scientific Name:
Notes:

TREE #8:
Common Name:
Scientific Name:
Notes:

TREE #9:
Common Name:
Scientific Name:
Notes:

TREE #10:
Common Name:
Scientific Name:
Notes:

TREE #11:
Common Name:
Scientific Name:
Notes:

TREE #12:
Common Name:
Scientific Name:
Notes:

TREE #13:
Common Name:
Scientific Name:
Notes:

TREE #14:
Common Name:
Scientific Name:
Notes:

FIELD BOTANY OPTION REQUIREMENT 8f1d:	TREE INVENTORY - Show two or more people how to use a leaf or twig key to identify at least five species of trees or shrubs.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

Notes:

FIELD BOTANY OPTION REQUIREMENT 8f2a:	TRANSECT STUDY - Visit two sites, at least one of which is different from the one you visited for Field Botany requirement 1.
FIELD BOTANY OPTION REQUIREMENT 8f2b:	TRANSECT STUDY - Use the transect method to study the two different kinds of plant communities. The transects should be at least 500 feet long.
FIELD BOTANY OPTION REQUIREMENT 8f2c:	TRANSECT STUDY - At each site, record observations about the soil and other influencing factors AND do the following. Then make a graph or chart to show the results of your studies.

SITE #1:

Place Name and Location:

Notes:

FIELD BOTANY OPTION SITE 1

REQUIREMENT 8f2c1: Identify each tree within 10 feet of the transect line.

REQUIREMENT 8f2c2: Measure the diameter of each tree at 4 feet above the ground, and map and list each tree

	Common Name	Diameter
Tree #1:		
Tree #2:		
Tree #3:		
Tree #4:		
Tree #5:		
Tree #6:		
Tree #7:		
Tree #8:		
Tree #9:		
Tree #10:		
Tree #11:		
Tree #12:		
Tree #13:		
Tree #14:		
Tree #15:		
Tree #16:		
Tree #17:		
Tree #18:		
Tree #19:		
Tree #20:		

Map of Trees:
SITE #2:
Place Name and Location:
Notes:

FIELD BOTANY OPTION SITE 2

REQUIREMENT 8f2c1: Identify each tree within 10 feet of the transect line.

REQUIREMENT 8f2c2: Measure the diameter of each tree at 4 feet above the ground, and map and list each tree

	Common Name	Diameter
Tree #1:		
Tree #2:		
Tree #3:		
Tree #4:		
Tree #5:		
Tree #6:		
Tree #7:		
Tree #8:		
Tree #9:		
Tree #10:		
Tree #11:		
Tree #12:		
Tree #13:		
Tree #14:		
Tree #15:		
Tree #16:		
Tree #17:		
Tree #18:		
Tree #19:		
Tree #20:		

Map of Trees:
Graph:

FIELD BOTANY OPTION REQUIREMENT 8f3a:	NESTED PLOT - Visit two sites, at least one of which is different from the one you visited for Field Botany requirement 1.
FIELD BOTANY OPTION REQUIREMENT 8f3b:	NESTED PLOT - Mark off nested plots and inventory two different kinds of plant communities.
FIELD BOTANY OPTION REQUIREMENT 8f3c:	NESTED PLOT - At each site, record observations about the soil and other influencing factors AND do the following. Then make a graph or chart to show the results of your studies. 1. Identify, measure, and map each tree in a 100-by-100-foot plot. (Measure the diameter of each tree larger than 3 inches in diameter at 4 feet above the ground.) 2. Identify and count all trees and shrubs in a 10-by-10-foot plot within each of the larger areas. 3. Identify and count all broad-leaved plants (trees, shrubs, vines, and herbaceous plants) all plants (wildflowers, ferns, grasses, mosses, etc.) of a 4-by-4-foot plot within the 10-by-10-foot plot.
SITE #1:	
Place Name and Location:	
Notes:	

FIELD BOTANY OPTION
REQUIREMENT 8f3c1:

SITE 1

Identify, measure, and map each tree in a 100-by-100-foot plot. (Measure the diameter of each tree larger than 3 inches in diameter at 4 feet above the ground.)

	Common Name	+3" Diameter
Tree #1:		
Tree #2:		
Tree #3:		
Tree #4:		
Tree #5:		
Tree #6:		
Tree #7:		
Tree #8:		
Tree #9:		
Tree #10:		
Tree #11:		
Tree #12:		
Tree #13:		
Tree #14:		
Tree #15:		
Tree #16:		
Tree #17:		
Tree #18:		
Tree #19:		
Tree #20:		
Tree #21:		

FIELD BOTANY OPTION
REQUIREMENT 8f3c2:

SITE 1

Identify and count all trees and shrubs in a 10-by-10-foot plot within each of the larger areas.

	Common Name	+3" Diameter
Tree / Shrub #1:		
Tree / Shrub #2:		
Tree / Shrub #3:		
Tree / Shrub #4:		
Tree / Shrub #5:		
Tree / Shrub #6:		
Tree / Shrub #7:		
Tree / Shrub #8:		
Tree / Shrub #9:		
Tree / Shrub #10:		
Tree / Shrub #11:		
Tree / Shrub #12:		
Tree / Shrub #13:		
Tree / Shrub #14:		
Tree / Shrub #15:		
Tree / Shrub #16:		
Tree / Shrub #17:		
Tree / Shrub #18:		
Tree / Shrub #19:		
Tree / Shrub #20:		
Tree / Shrub #21:		

FIELD BOTANY OPTION REQUIREMENT 8f3c3:	SITE 1 Identify and count all broad-leaved plants (trees, shrubs, vines, and herbaceous plants) all plants (wildflowers, ferns, grasses, mosses, etc.) of a 4-by-4-foot plot within the 10-by-10-foot plot.

PLANT #1 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #2 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #3 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #4 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #5 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #6 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #7 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #8 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #9 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #10 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #11 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #12 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐

SITE #2:
Place Name and Location:
Notes:

FIELD BOTANY OPTION REQUIREMENT 8f3c1: SITE 2

Identify, measure, and map each tree in a 100-by-100-foot plot. (Measure the diameter of each tree larger than 3 inches in diameter at 4 feet above the ground.)

	Common Name	+3" Diameter
Tree #1:		
Tree #2:		
Tree #3:		
Tree #4:		
Tree #5:		
Tree #6:		
Tree #7:		
Tree #8:		
Tree #9:		
Tree #10:		
Tree #11:		
Tree #12:		
Tree #13:		
Tree #14:		
Tree #15:		
Tree #16:		
Tree #17:		
Tree #18:		
Tree #19:		
Tree #20:		
Tree #21:		

FIELD BOTANY OPTION
REQUIREMENT 8f3c2:

SITE 2

Identify and count all trees and shrubs in a 10-by-10-foot plot within each of the larger areas.

	Common Name	+3" Diameter
Tree / Shrub #1:		
Tree / Shrub #2:		
Tree / Shrub #3:		
Tree / Shrub #4:		
Tree / Shrub #5:		
Tree / Shrub #6:		
Tree / Shrub #7:		
Tree / Shrub #8:		
Tree / Shrub #9:		
Tree / Shrub #10:		
Tree / Shrub #11:		
Tree / Shrub #12:		
Tree / Shrub #13:		
Tree / Shrub #14:		
Tree / Shrub #15:		
Tree / Shrub #16:		
Tree / Shrub #17:		
Tree / Shrub #18:		
Tree / Shrub #19:		
Tree / Shrub #20:		
Tree / Shrub #21:		

FIELD BOTANY OPTION REQUIREMENT 8f3c3:	SITE 2 Identify and count all broad-leaved plants (trees, shrubs, vines, and herbaceous plants) all plants (wildflowers, ferns, grasses, mosses, etc.) of a 4-by-4-foot plot within the 10-by-10-foot plot.

PLANT #1 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #2 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #3 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #4 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #5 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #6 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #7 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #8 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #9 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #10 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #11 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐
PLANT #12 Name:	WILDFLOWER ☐	FERN ☐	GRASS ☐	MOSS ☐	OTHER ☐

Graph:

FIELD BOTANY OPTION REQUIREMENT 8f4a:	HERBARIUM VISIT - Write ahead and arrange to visit an herbarium at a university, park, or botanical garden; OR, visit an herbarium website (with your parent or guardian's permission).

PARENT / GUARDIAN PERMISSION: IS REQUIRED.

Parent's / Guardian's Name _____ Phone or Email _____

Parent's / Guardian's Signature _____ Date _____ ☐ *permission*

Place Name and Location and website url:

FIELD BOTANY OPTION REQUIREMENT 8f4b:	HERBARIUM VISIT - Tell how the specimens are arranged.

Notes:

FIELD BOTANY OPTION REQUIREMENT 8f4b:	HERBARIUM VISIT - Tell how the specimens are used by researchers.

Notes:

FIELD BOTANY OPTION REQUIREMENT 8f4b:	HERBARIUM VISIT - If possible, observe voucher specimens of a plant that is rare in your state.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

FIELD BOTANY OPTION REQUIREMENT 8f4c:	HERBARIUM VISIT - Tell how a voucher specimen is mounted and prepared for permanent storage.
Notes:	
FIELD BOTANY OPTION REQUIREMENT 8f4c:	HERBARIUM VISIT - Tell how specimens should be handled so that they will not be damaged.
Notes:	
FIELD BOTANY OPTION REQUIREMENT 8f4d:	HERBARIUM VISIT - Tell about the tools and references used by botanists in an herbarium.
Notes:	

FIELD BOTANY OPTION REQUIREMENT 8f5a:	PLANT CONSERVATION ORGANIZATION VISIT - Write ahead and arrange to visit a private conservation organization or government agency that is concerned with protecting rare plants and natural areas.
Place Name and Location:	
Notes:	
FIELD BOTANY OPTION REQUIREMENT 8f5b:	PLANT CONSERVATION ORGANIZATION VISIT - Tell about the activities of the organization in studying and protecting rare plants and natural areas.
Notes:	

FIELD BOTANY OPTION REQUIREMENT 8f5c:	PLANT CONSERVATION ORGANIZATION VISIT - If possible, visit a nature preserve managed by the organization. Tell about land management activities such as controlled burning, or measures to eradicate invasive (nonnative) plants or other threats to the plants that are native to the area.
Place Name and Location:	
Notes:	